BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MALAKAND

INTERMEDIATE (12TH CLASS) EXAMINATION, 2025

CHEMISTRY

Past Paper 2025 • Paper Code: 1

Retyped faithfully from the original board paper — same questions, no changes.

SECTION "A"

Time Allowed: 20 Minutes | **Maximum Marks:** 18 | **Paper Code:** 1

Note: There are Three Sections in this paper i.e. A, B and C. Attempt Section-A and return it to the Superintendent within the given time. No marks will be awarded for Cutting, Erasing and Overwriting. Mobile Phone is strictly prohibited in Examination Hall. Q.1. Choose the correct option and shade one of (A, B, C, D) in the given Bubble Answer Sheet.

Q1. A metal with completely filled or empty d-orbitals is:

(A) Ferromagnetic
(B) Paramagnetic
(C) Colored
(D) Colorless

Q2. Down the group in periodic table the reducing property of alkali metals:

(A) Decreases
(B) Increases
(C) Constant
(D) Zero

Q3. Benzene gives addition product when reacts with:

(A) Cl_2 + UV
(B) Cl_2 + $FeCl_3$
(C) HNO_3 + H_2SO_4
(D) Fuming H_2SO_4

Q4. Self-linking property of carbon atoms is:

(A) Isomerization
(B) Aromaticity
(C) Catenation
(D) Carbonation

Q5. The relatively more stable alkene is:

(A) Isobutylene
(B) 1-butene
(C) Trans-2-butene
(D) Cis-2-butene

Q6. The most reactive alkyl halide is:

(A) R-F
(B) R-Cl
(C) R-Br
(D) R-I

Q7. An amphoteric oxide of the following is:

(A) MgO
(B) Na_2O
(C) SiO_2
(D) Al_2O_3

Q8. Ozonolysis of 2-butene produces:

(A) Acetone
(B) Butanone
(C) Formaldehyde
(D) Acetaldehyde

Q9. All the protons are equivalent in:

(A) CH_3OH
(B) CH_3NH_2
(C) C_2H_5OH
(D) $(CH_3)_4Si$

Q10. Bimolecular elimination reaction of alkyl halides is:

(A) Zero order
(B) 1st order
(C) 2nd order
(D) 3rd order

Q11. Delayed wound healing is caused by deficiency of:

(A) Zinc
(B) Iron
(C) Calcium
(D) Manganese

Q12. Which compound forms red precipitate with Fehling's solution?

(A) Acetone
(B) Acetaldehyde
(C) Butanone
(D) Acetophenone

Q13. Oxidation of primary alcohols produces:

(A) Aldehydes
(B) Alkynes
(C) Alkenes
(D) Ketones

Q14. The coldest region of atmosphere is:

(A) Troposphere
(B) Stratosphere
(C) Mesosphere
(D) Exosphere

Q15. The rate of SN^1 reaction of alkyl halides is independent of the nature of:

(A) Substrate
(B) Nucleophile
(C) Solvent
(D) Leaving group

Q16. The NMR active specie of the following is:

(A) C^{12} (B) N^{15}

(C) O^{16} (D) S^{32}

Q17. The more basic in aqueous solution is:

(A) NH_3 (B) CH_3NH_2

(C) $(CH_3)_2NH$ (D) $(CH_3)_3N$

Q18. A halogen of the highest electron affinity is:

(A) Fluorine (B) Chlorine

(C) Bromine (D) Iodine

SECTION "B"

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 40

Q2. Give Answers to any TEN (10) questions of the following. All questions carry equal marks.

i. Why compounds of SN+4 are covalent and SN+2 are ionic in nature?

ii. What are addition and condensation polymers?

iii. How and why binding energy of transition metals varies across the period?

iv. What is photochemical smog? How it is formed?

v. How E1 and E2 mechanisms differ from each other?

vi. Why nitrates and carbonates of group-IIA metals are thermally stable?

vii. What is biological oxygen demand and chemical oxygen demand?

viii. What are the features of a homologous series?

ix. Why bond enthalpy and electron affinity of chlorine is higher than fluorine?

x. How halogens are o/p-directing but deactivating?

xi. How carboxylic acids can be converted into acid amides and acid anhydrides?

xii. What is the significance of finger print region and functional group region?

xiii. Write the two steps in SN1 mechanism of alkyl halides.

SECTION "C"

Maximum Marks: 27

> Note: Attempt any THREE (03) questions from the following. Each question carries equal marks.

Q3. (Marks 09) Discuss the solubility of hydroxides, sulphates and carbonates of alkaline earth metals.

Q4. (Marks 09) What is stereoisomerism? Discuss in detail optical and geometrical isomerism.

Q5. (Marks 09) What are amines? Discuss the basicity and chemical reactions of amines with aldehydes, ketones and acid halides.

Q6. (Marks 09) Discuss the different methods of preparation of alcohols.